第一章 作业.

A组.

1. (1) $A \cap \bar{B} \cap \bar{C}$ (2) $A\bar{B}\bar{C} \cup \bar{A}\bar{B}C \cup \bar{A}B\bar{C}$ (3) $\overline{\bar{A}\bar{B}\bar{C}}$ (4) $\bar{A}\bar{B}C \cup A\bar{B}\bar{C} \cup \bar{A}\bar{B}C \cup \bar{A}B\bar{C}$

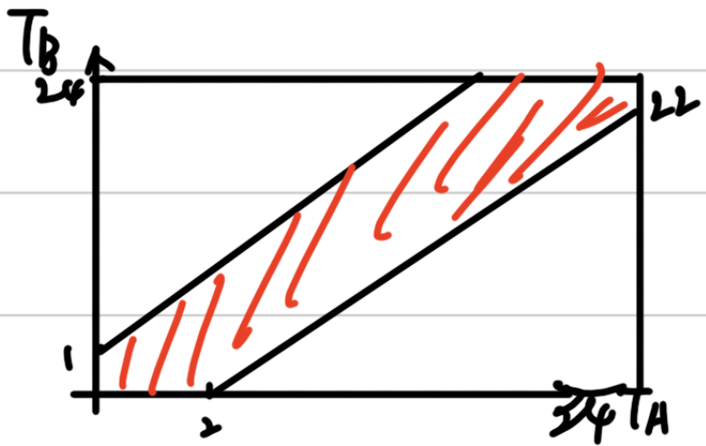
(5) $\bar{A}\bar{B}\bar{C}$ (6) $\bar{A}BC \cup \bar{A}\bar{B}C \cup \bar{A}B\bar{C}$

3. (1) $\{x \mid 1 \le x \le 4\}$ (2) $\{x \mid 2 < x \le 3\}$ (3) $\{x \mid 0 \le x < 1\} \cup \{x \mid 3 < x \le 5\}$ (4) $\{x \mid 1 \le x \le 2\}$

5. 设事件A为"正方体有2面上色".

$$P(A) = \frac{96}{1000} = 9.6\%$$

7.



i. 若A船先到. $T_B - T_A < 1$

ii. 若B先到. $T_A - T_B < 2$

$$p = \frac{24\times24 - \frac{1}{2}\times12^2 - \frac{1}{2}\times12^2}{24\times24} \approx 12.07\%$$

9. (1) 因 $P(AC)=0$, 故 $P(ABC)=0$.

(2). $P(A\cup B\cup C) = P(A)+P(B)+P(C)-P(AB)-P(BC)-P(AC)+P(ABC)$

$= \frac{1}{2}$

(3) $P(\bar{A}\bar{B}\bar{C}) = 1 - P(A\cup B\cup C) = \frac{1}{2}$

11. (1) 取后不放回 $P(A) = \frac{4}{10}\cdot\frac{3}{9} = \frac{2}{15}$ $P(B) = \frac{4}{10}\times\frac{6}{9} = \frac{4}{15}$ $P(C) = C_2^1\times\frac{4}{10}\times\frac{6}{9} = \frac{8}{15}$ $P(D) = \frac{2}{5}$

(2) 有回. $P(A) = \frac{4}{10}\times\frac{4}{10} = \frac{4}{25}$ $P(B) = \frac{4}{10}\times\frac{6}{10} = \frac{6}{25}$ $P(C) = C_2^1\times\frac{4}{10}\times\frac{6}{10} = \frac{12}{25}$ $P(D) = \frac{2}{5}$

13. $P(\text{至少2人同}) = 1 - P(\text{各不同}) = 1 - \frac{A_{12}^4}{12^4} = \frac{41}{96}$

15. 末位0: $9\times8\times7 = 504$.

末位偶：$4\times8\times8\times1=128$　　$P=\frac{504+128}{5040}=\frac{41}{90}$

17. $P(A\cup B)=P(A)+P(B)-P(AB)=\frac{1}{12}$

$P(\bar{C}\bar{B}\mid A\cup B)=\frac{P(A\bar{B})}{P(A\cup B)}=\frac{P(A)-P(AB)}{\frac{1}{12}}=\frac{5}{9}$

19. $P=\frac{7\times6\times93}{100\times99\times98}=\frac{31}{7700}$

21. $P=\frac{4}{20}\times0.9+\frac{8}{20}\times0.7+\frac{7}{20}\times0.5+\frac{1}{20}\times0.2=0.645$

23.(1) $P=\frac{4}{9}\times0.8+\frac{2}{9}\times0.6+\frac{2}{9}\times0.7=\frac{32}{45}$

(2) 设 G_1 为"好评".

$P(B\mid G_1)=\frac{P(B)P(G_1\mid B)}{P(G_1)}=\frac{9}{32}$

33. 设 $P(A|B) = P(A|\bar{B}) = c$. $P(A) = P(A|B)P(B) + P(A|\bar{B})P(\bar{B}) = c(P(B) + P(\bar{B})) = c$

∴ $P(AB) = P(A|B)P(B) = cP(B) = P(A)P(B)$ ∴ 独立.

34. $P = 1 - P(\text{均未击中}) = 1 - (1-\frac{4}{5})(1-\frac{2}{3})(1-\frac{3}{4}) = \frac{59}{60}$

29. $P(I) = 1 - (1-p^n)^2 = 2p^n - p^{2n}$

II中某组: $1-(1-p)^2 = 2p - p^2$, $P(II) = (2p-p^2)^n > P(I)$

∴ II更可靠.

直观来讲，若I工作，则有个上下组必有一个正常. 此时II必工作.

B组.

1.(1) $P=\dfrac{2^5\times A_5^5}{A_{10}^{10}}=\dfrac{1}{945}$

(2) $P=\dfrac{1}{2^5}=\dfrac{1}{32}$

(3) 一圈共 $(10-1)!=9!$ 种. $P=\dfrac{2^3\cdot 4!}{9!}=\dfrac{2}{945}$

3. (1) $P_{甲}=0.25\times(3\times0.2\times0.8^2)+0.6\times(3\times0.2^2\times0.8)+0.15\times0.2^3=0.1612$

(2) $P=\dfrac{0.6\times(3\times0.2^2\times0.8)}{P_{甲}}=\dfrac{144}{403}$

5. (1) $P_1=p\cdot[1-(1-p)^2]=2p^3-p^5$

$P_2=(1-p)\cdot[1-(1-p)^2]+p\cdot[1-(1-p)^2]^2$

5次小 5次.

$= 2p^2 + 2p^3 - 5p^4 + 4p^5$

(2) 设事件 E "1正等1负", S "1,2均出现".

$\therefore P(ES) = C(1-p^2)p^3$

$P(E|S) = \frac{P(ES)}{P(S)} = \frac{C(1-p^2)p^3}{2p^3-p^5} = \frac{(1-p)^2}{2-p^2}$